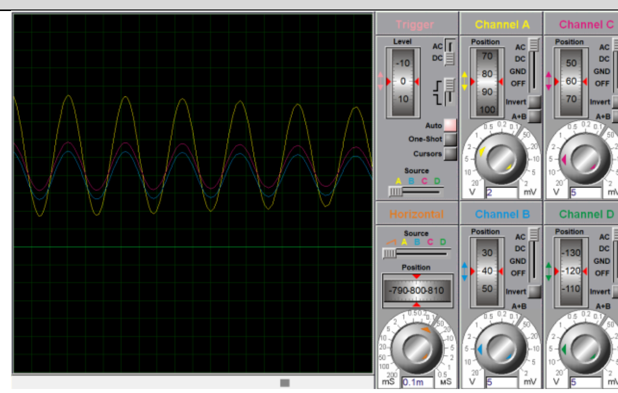
Pertemuan 14

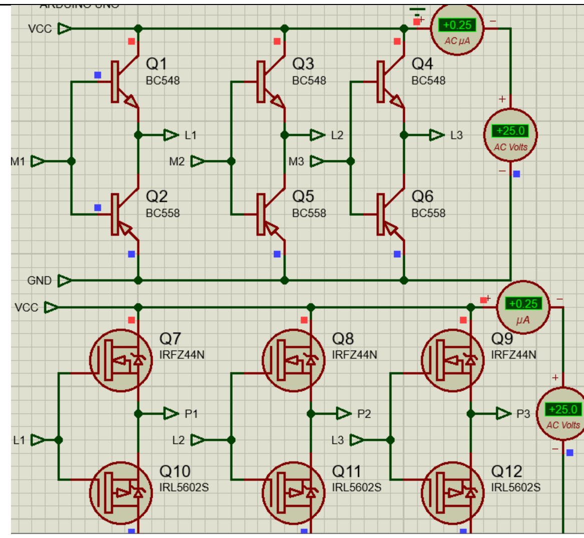
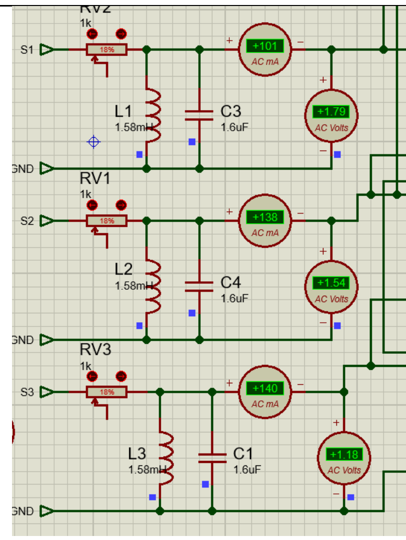
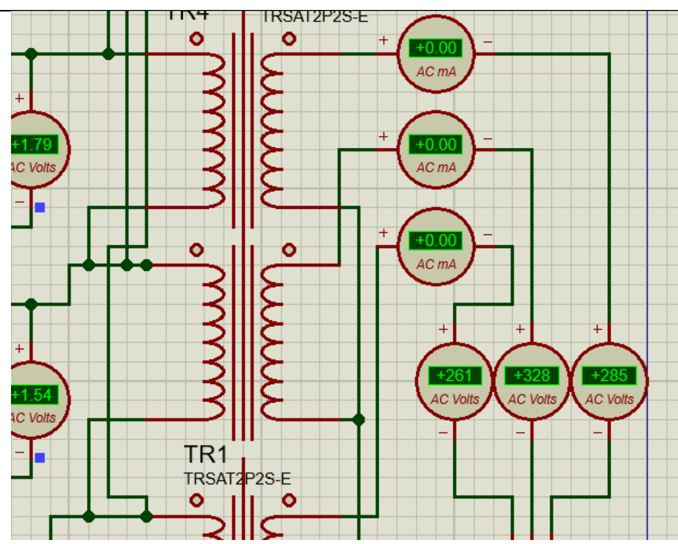
1. Pembagian Ukuran Frekuensi

NO	Nama	Penentuan Frekuensi (Hz)
1	Grimonia Fredlina Marceli	20-50
2	Meira Sukmahayu Nareswari	30-60
3	Ariska Meyny Ningrum	40-70
4	Marshanda Ifany Ashari	50-80
5	Rizq Syandana Dwi Anggara	60-90
6	Muhandis Lawdza'i Putra Sanjaya	70-100
7	Fuad Galih Pambudi	80-110
8	Delia Nurhidayah	90-120
9	Putri Nilam Sari	100-130
10	RANGGA IVAN NUGRAHA	110-140
11	DA'I SABILI WAHYU KUSUMO	120-150
12	Satria Bhekti Akhisnu	130-160
13	Kristina Wulandari	140-170

2. Progres Proyek

1) Nilai keluaran pada setiap komponen (Arduino, Transistor/Mosfet, Filter (RLC), Trafo, Motor). Nilai tersebut berupa tegangan dan gelombang

NO	Komponen	Tegangan	Gelombang
1	Arduino	25V	

<table>
<tr><td>2</td><td>Transistor atau Mosfet</td><td>25V</td><td>
</td></tr>
<tr><td>3</td><td>Filter (RLC)</td><td>1.50V</td><td>
</td></tr>
<tr><td>4</td><td>Trafo</td><td>315V</td><td>
</td></tr>
<tr><td>5</td><td>Motor</td><td>0V</td><td></td></tr>
</table>

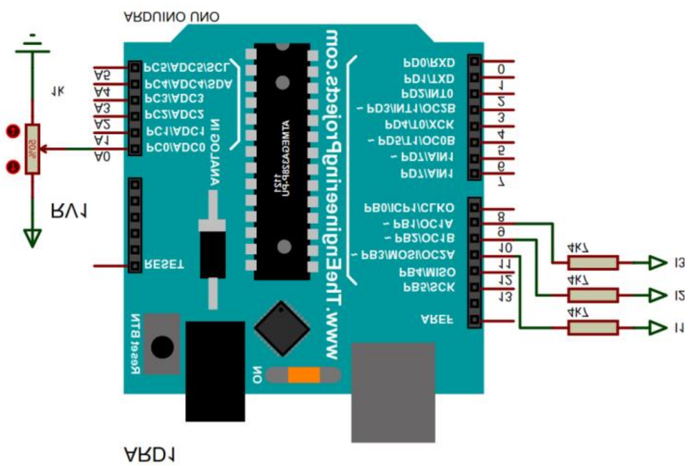
2) Perhitungan ukuran variable komponen (Frekuensi input, Filter (RLC), Trafo).

NO	Frekuensi Input	Filter (RLC)	Trafo
1	1/F/3 x 1.000.000	R (1k) C (1.6uF) $L = \frac{1}{(2\pi f_0)^2 C}$ L (1,58 mH)	0.04

3) Progres pengukuran dimasukan di lembar pengesahan/Kesimpulan

4) Contoh penamnahan rangkaian dan program untuk resistor variable perubahan frekuensi :

   a. Rangkaian



   b. Program

```
int val;
void setup(){
  pinMode(9, OUTPUT); // Phase 1
  pinMode(10, OUTPUT); // Phase 2
  pinMode(11, OUTPUT); // Phase 3
}
void loop(){
  val = analogRead(0);
  val = map(val, 0, 1023, Ton1, Ton2); //Ton1=batas bawah F //Ton2=batas atas F
  delayMicroseconds(val);
  digitalWrite(9, LOW);
  delayMicroseconds(val);
```

```
  digitalWrite(10, HIGH);
  delayMicroseconds(val);
  digitalWrite(11, LOW);
  delayMicroseconds(val);
  digitalWrite(9, HIGH);
  delayMicroseconds(val);
  digitalWrite(10, LOW);
  delayMicroseconds(val);
  digitalWrite(11, HIGH);
}
```